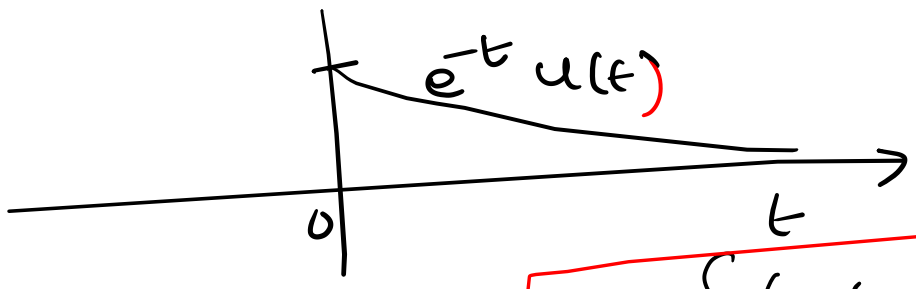
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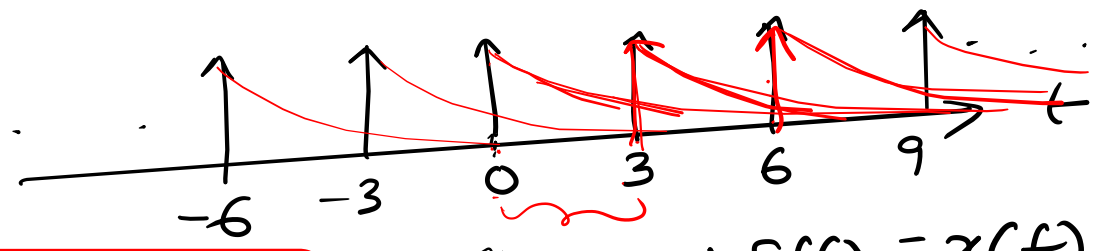
$$y(t) = \left(e^{-t}u(t)\right) * \sum_{k=-\infty}^{\infty} \delta(t-3k)$$

T.S.T. $y(t) = A e^{-t}$ for $0 \le t < 3$ & determine the value of A.

$$y(t) = x_1(t) * x_2(t)$$

$$x_1(t) = e^{-t}u(t) \quad \& \quad x_2(t) = \sum_{k=-\infty}^{\infty} \delta(t-3k)$$





$$y(t) = \sum_{k=-\infty}^{\infty} \left\{ \left(e^{-t}u(t)\right) * \delta(t-3k) \right\}$$

$x(t) * \delta(t) = x(t)$

$x(t) * \delta(t-t_0) = x(t-t_0)$

✓

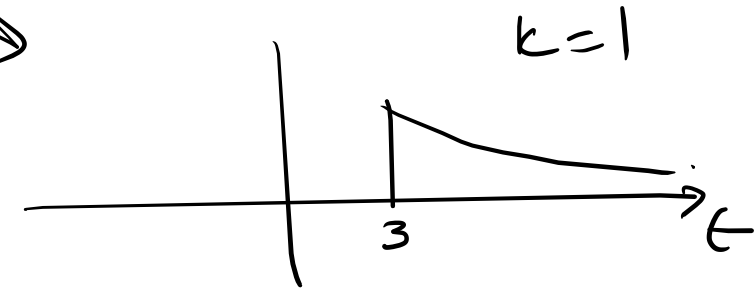
$k=0$: $e^{-t}u(t) * \delta(t) = e^{-t}u(t)$



X

$k=1$: $e^{-t}u(t) * \delta(t-3)$

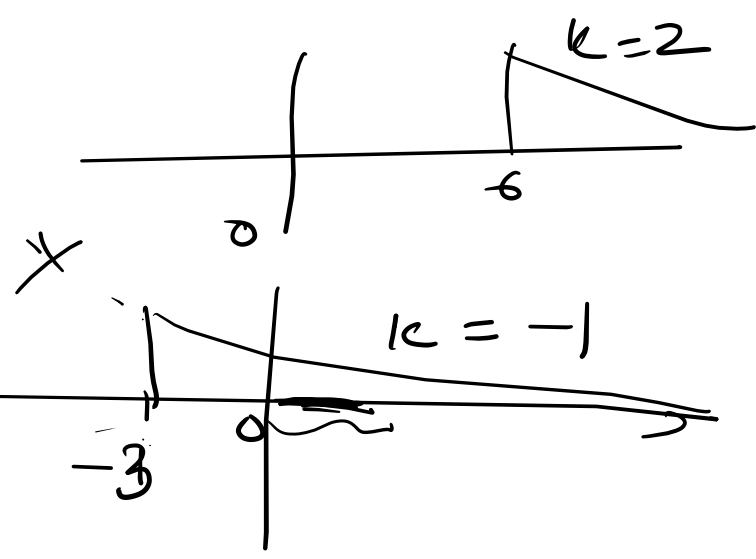
$= e^{-(t-3)}u(t-3)$



$0 \le t < 3$

X

$k=2$: $e^{-t}u(t) * \delta(t-6)$



X

$k=-1$: $e^{-t}u(t) * \delta(t+3)$

1 $y(t)$ = If $0 \le t < 3$

$$= \sum_{k=-\infty}^{0} \left(e^{-t} u(t) \right) * \left(\delta(t-3k) \right)$$

$y(t) = Ae^{-t} \quad 0 \le t < 3$

$$= \sum_{k=-\infty}^{0} e^{-(t-3k)} u(t-3k)$$

$$A = \frac{1}{1-e^{-3}}$$

$$= \underset{\uparrow k=0}{e^{-t}} + \underset{\uparrow k=-1}{e^{-(t+3)}} + \underset{\uparrow k=-2}{e^{-(t+6)}} + \cdots$$

$$= e^{-t}\left[1 + e^{-3} + e^{-6} + \cdots \right]$$

$$y(t) = \frac{e^{-t}}{1-e^{-3}} \qquad 0 \le t < 3$$

LTI systems $\Rightarrow$ completely characterized by its impulse response.

S_1: $y[n] = x[n] + x[n-1]$

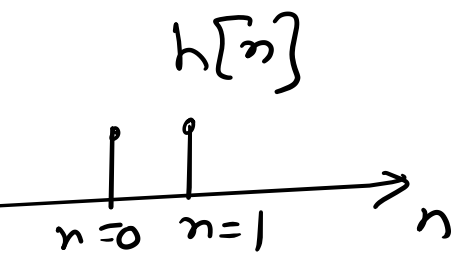
↓

System is unique given an i/p - o/p behaviour

impulse response of the system —

if $x[n] = \delta[n]$ $\Rightarrow$ $y[n] = h[n]$

$h[n] = \delta[n] + \delta[n-1]$



if $x[n] = \delta[n-1]$ $\Rightarrow$ $y[n] = h[n-1]$

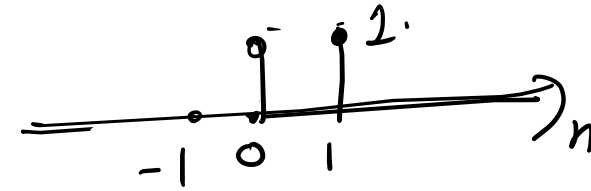
if $x[n] = \alpha_1 \delta[n-1] + \alpha_2 \delta[n-2]$ $\Rightarrow$ $y[n] = \alpha_1 h[n-1] + \alpha_2 h[n-2]$

$S_2 : y[n] = (x[n] + x[n-1])^2$

$h[n]$

if $x[n] = \delta[n] \Rightarrow y[n] = (\delta[n] + \delta[n-1])^2$

$\updownarrow \quad \equiv$



$x[n] = \alpha_1 \delta[n-1] + \alpha_2 \delta[n-2]$

$y[n] \neq \alpha_1 h[n-1] + \alpha_2 h[n-2]$

---

if $x[n] = \alpha_1 \delta[n-1] \Rightarrow y[n] = \alpha_1^2 (\delta[n-1] + \delta[n-2])^2$

$\neq \alpha_1 h[n-1]$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

① if LTI system is memoryless
then $h[n] = k\delta[n]$ & $h(t) = k\delta(t)$

---

② if LTI system is causal, then
$h[n] = 0 \quad n < 0$ & $h(t) = 0 \quad t < 0$

Let $h[n] = \delta[n-3]$

System with memory

$y[n] = x[n] * h[n] = x[n] * \delta[n-3] = x[n-3]$

$y[0] = x[-3]$

LTI system is causal —

D.T.

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} x[k]\, h[n-k] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

if system is causal, then o/p depends on present & past i/ps.

$$\Rightarrow y[n] = \sum_{k=-\infty}^{n} x[k]\, h[n-k]$$

$n-k=r \Rightarrow$ if $k=-\infty \Rightarrow r=\infty$

if $k=n \Rightarrow r=0$

$$y[n] = \sum_{r=0}^{\infty} x[n-r]\, h[r] = \sum_{r=0}^{\infty} h[r]\, x[n-r]$$

$h[r] = 0 \quad r<0$ or $h[n]=0 \quad n<0$

C.T.

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau)\, d\tau$$

System is causal,

$$y(t) = \int_{-\infty}^{t} x(\tau) h(t-\tau)\, d\tau$$

$$= \int_{0}^{\infty} h(t)\, x(t-\tau)\, d\tau$$

$h(t) = 0 \quad ; t<0$

$h[n]$

③ BIBO Stability

A Stable System with Bounded i/p generates a bounded o/p

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$= \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

Let i/p is bounded for all n

$$|x[n]| < M \quad \forall n$$

where M is a finite no.

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} h[k]\, x[n-k] \right|$$

$$\leq \sum_{k=-\infty}^{\infty} |h[k]|\, |x[n-k]|$$

$$\leq M \sum_{k=-\infty}^{\infty} |h[k]|$$

if $y[n]$ is to be bounded, this implies that

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

$\Rightarrow$ if $h[k]$ is absolutely summable, then the System is stable

$\therefore$ it will generated bounded o/p for bounded i/p.

For a C.T. system

a system is stable if it is absolutely integrable.

i.e) $\int_{-\infty}^{\infty} |h(t)|\, dt < \infty$

Let $h(t) = e^{-t} u(t)$

— memoryless $\Rightarrow$ NO.

→ causality $\Rightarrow$ Yes.

→ stability $\Rightarrow$ Yes.

$$\int_{-\infty}^{\infty} e^{-t} u(t)\, dt$$

$$= \int_{0}^{\infty} e^{-t} dt = \frac{e^{-t}}{-1}\Big|_0^{\infty}$$

$$= 1$$